ModelIt!

Student Activity Pack

Wave Properties: Amplitude, Frequency, and Speed

How Energy and Medium Shape Every Wave in the Universe

Grade 08L2 | NGSS MS-PS4-1, MS-PS4-2 | Patterns

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is a wave, and can you name different types of waves?

2 What is the difference between a loud sound and a quiet sound at the wave level?

3 Does sound travel faster in air or water? Why do you think so?

4 Draw a wave and label any properties you know (amplitude, wavelength, frequency).

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wave Source Energy | Medium Type | Amplitude | Frequency | Wavelength

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Quiet vs. Loud

Low energy vs. high energy in air

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Air to Water

Same source energy, switch medium from air to water

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High Frequency

High source energy in different media

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Energy Carried — The total energy transported by the wave, related to both amplitude and frequency

Reflection Percentage — How much of the wave bounces back when it hits a boundary between media

Absorption Rate — How much wave energy is converted to heat as it passes through a medium

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does a whisper sound different from a shout, and why does sound travel faster underwater?
Use evidence from your simulation to support your answers.

1 Why does sound seem different underwater compared to in air?

2 How does the inverse relationship between frequency and wavelength help explain why radio waves are long but gamma rays are short?

3 Why does frequency stay the same when a wave enters a new medium, but speed and wavelength change?

4 How does amplitude relate to the energy of a wave? What real-world evidence supports this?

5 How could understanding wave properties help in designing earthquake-resistant buildings?

8 STEM Challenge: Design a Sonar Detection System

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YOUR ENGINEERING MISSION

Design a sonar system that uses wave properties to detect underwater objects at different depths, using your model data to select the optimal frequency and energy settings.

THE SCENARIO

A marine research team needs a sonar system for their submarine. They need to detect objects at various depths in the ocean. Your team must determine the best wave frequency, amplitude, and medium considerations for their detection system.

Engineering Design Thinking

1. WHAT WAVE FREQUENCY WILL YOUR SYSTEM USE AND WHY?
2. HOW WILL AMPLITUDE AFFECT DETECTION RANGE?
3. HOW WILL YOU ACCOUNT FOR WAVE SPEED CHANGES IN DIFFERENT WATER TEMPERATURES?
4. HOW WILL YOU CALCULATE THE DISTANCE TO A DETECTED OBJECT USING WAVE SPEED AND RETURN TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Acoustical Engineers and Wave Physicists design systems that use wave properties for applications ranging from medical ultrasound to earthquake detection to concert hall design. They earn $70,000-$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Your model shows that when a sound wave moves from air (343 m/s) to water (1,480 m/s), wave speed increases but frequency stays the same. Using the wave equation v = f x wavelength, what must happen to wavelength?

- **A.** Wavelength decreases because higher speed compresses the waves
- **B.** Wavelength increases proportionally because speed increased while frequency remained constant
- **C.** Wavelength stays the same because it is a fixed property of the wave
- **D.** Wavelength becomes zero because the wave cannot exist in water

2 In the model, increasing Wave Source Energy from 20% to 90% increased Amplitude dramatically. A student asks: does doubling the amplitude double the energy? What does the model reveal?

- **A.** Yes, amplitude and energy have a direct 1:1 relationship
- **B.** No, energy is proportional to the square of amplitude. Doubling amplitude quadruples the energy carried by the wave.
- **C.** No, amplitude has no relationship to energy
- **D.** Yes, but only for sound waves, not light waves

3 Your model shows a NEGATIVE relationship between Frequency and Wavelength. Which real-world example best illustrates this inverse relationship?

- **A.** Radio waves have low frequency and long wavelength, while gamma rays have high frequency and short wavelength
- **B.** Loud sounds have long wavelengths and quiet sounds have short wavelengths
- **C.** All waves have the same frequency and wavelength regardless of type
- **D.** Higher frequency waves always travel slower than lower frequency waves

4 A student observes that frequency stays the same when a wave enters a new medium, even though speed and wavelength change. Why does frequency remain constant?

- **A.** Because frequency is a universal constant that never changes for any wave
- **B.** Because frequency is determined by the source, which continues vibrating at the same rate regardless of what medium the wave enters
- **C.** Because the new medium forces the frequency to match the old medium
- **D.** Because frequency and speed are the same property measured differently

5 **Based on your model, a marine scientist needs sonar to detect small objects on the ocean floor. High-frequency sonar provides fine detail but has short range, while low-frequency sonar travels far but provides less detail. Which model evidence supports choosing high-frequency sonar for this task?**

- **A.** High-frequency waves are louder and can bounce off smaller objects
- **B.** High-frequency waves have shorter wavelengths, which can resolve smaller features because the wavelength is closer to the size of the objects being detected
- **C.** High-frequency waves travel faster in water, reaching the ocean floor sooner
- **D.** High-frequency waves carry less energy, making them safer for marine life